

Brian K. Kirui

POSTDOCTORAL RESEARCHER · EPIDEMIOLOGY

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Education

University of Gothenburg

PH.D. EPIDEMIOLOGY

Gothenburg, Sweden

2022–2026

University of Gothenburg

MASTER OF MEDICAL SCIENCE (MMSc) IN GLOBAL HEALTH

Gothenburg, Sweden

2019–2021

Pharmacy and Poisons Board

LICENSED AND REGISTERED PHARMACIST (RPH), KENYA

Nairobi, Kenya

2018

Kenyatta University

BACHELOR OF PHARMACY (BPHARM)

Nairobi, Kenya

2011–2016

Research Experience

Department of Public Health and Clinical Medicine, The OLIN and Sunderby Research

Unit, Umeå University

Umeå, Sweden

POSTDOCTORAL RESEARCHER, OBSTRUCTIVE LUNG DISEASES

2026 – Present

- Conduct epidemiological research on obstructive lung diseases using population-based cohorts and healthcare registers.
- Develop and lead studies investigating disease progression, multimorbidity, and long-term health outcomes.
- Collaborate with national and international research groups on respiratory and public health research initiatives.

School of Public Health and Community Medicine, University of Gothenburg

Gothenburg, Sweden

DOCTORAL STUDENT, EPIDEMIOLOGY

May 2022 – Present

- Investigating risk factors and long-term outcomes of COVID-19 among individuals with obstructive lung disease using Swedish population registers.
- Conduct epidemiological and register-based studies utilizing nationwide healthcare and demographic databases.
- Perform advanced statistical analyses and contribute to scientific publications in respiratory and public health research.

School of Public Health and Community Medicine, University of Gothenburg

Gothenburg, Sweden

ASSOCIATE RESEARCHER

Aug 2021 – Feb 2022

- Supported studies examining COVID-19 outcomes among patients with asthma and chronic obstructive pulmonary disease.
- Contributed to study design, statistical analysis plans, implementation, reporting, and research documentation.

School of Public Health and Community Medicine, University of Gothenburg

Gothenburg, Sweden

MASTER'S THESIS RESEARCHER

Jan 2021 – Jun 2021

- Conducted register-based research on epidemiological risk factors for COVID-19 among individuals with asthma and COPD in Sweden.
- Applied epidemiological and statistical methods using Swedish population-based health registers.

School of Public Health and Community Medicine, University of Gothenburg

Gothenburg, Sweden

RESEARCH INTERN

Nov 2020 – Jan 2021

- Completed advanced training in Swedish register-based research and statistical analysis.
- Described characteristics of individuals with pre-existing respiratory conditions who contracted COVID-19 using national register data.

Publications

* Co-first author.

PEER-REVIEWED PUBLICATIONS

1. Jang SY, Hussain-Alkhateeb L, Rivera Ramirez T, Al-Aghbari AA, Chackalackal DJ, Cardenas-Sanchez R, Carrillo MA, Oh I-H, Alfonso-Sierra EA, Oechsner P, **Kirui BK**, Anto M, Diaz-Monsalve S, Kroeger A. Factors shaping the COVID-19 epidemic curve: a multi-country analysis. *BMC Infectious Diseases*. 2021;21(1):1032.
2. Nyberg F, Lindh M, Vanfleteren L, Hammar N, Wettermark B, Sundström J, Santosa A, **Kirui BK**, Gisslén M. Adverse events of special interest for COVID-19 vaccines – background incidences vary by sex, age and time period and are affected by the pandemic. *medRxiv*. 2021.
3. Xu Y, Li H, **Kirui BK**, Santosa A, Gisslén M, Leach S, Wettermark B, Vanfleteren LEGW, Nyberg F. Effectiveness of COVID-19 Vaccines over 13 Months Covering the Period of the Emergence of the Omicron Variant in the Swedish Population. *Vaccines*. 2022;10(12):2074.
4. **Kirui BK**, Santosa A, Vanfleteren LEGW, Li H, Franzén S, Stridsman C, Nyberg F. Pre- and Post-vaccination Characteristics and Risk Factors for COVID-19 Outcomes in a Swedish Population-Based Cohort of COPD Patients. *ERJ Open Research*. 2023;9(2).
5. Labor M, **Kirui BK**, Nyberg F, Vanfleteren LEGW. Regular Inhaled Corticosteroids Use May Protect Against Severe COVID-19 Outcome in COPD. *International Journal of Chronic Obstructive Pulmonary Disease*. 2023;18:1701–1712.
6. Einarsdottir MJ*, **Kirui BK***, Li H, Olsson D, Johannsson G, Nyberg F, Ragnarsson O. Impact of Chronic Oral Glucocorticoid Treatment on Mortality in Patients with COVID-19: Analysis of a Population-Based Cohort. *BMJ Open*. 2024;14(3):e080640.
7. Santosa A, Oras J, Li H, Nwaru C, **Kirui BK**, Nyberg F. Survival of Critically Ill COVID-19 Patients in Sweden During the First Two and a Half Years of the Pandemic. *Critical Care Medicine*. 2024.
8. **Kirui BK**, Santosa A, Li H, Vanfleteren LEGW, Stridsman C, Nyberg F. Key Characteristics of Asthma Patients with COVID-19 Vary Substantially by Age. *Journal of Asthma and Allergy*. 2024;17:589–600.
9. Li H, Xu Y, **Kirui BK**, Santosa A, Nyberg F. Ursodeoxycholic Acid Treatment Did Not Show Protective Effect for Severe COVID-19 Outcomes – A Nationwide Register Study. *BMC Public Health*. 2026;26(1):1240.
10. **Kirui BK**, Wallström W, Vanfleteren LEGW, Li H, Santosa A, Bonander C, Stridsman C, Börjesson M, Nyberg F. Cardiovascular and Mortality Risks Following COVID-19-Related vs. Non-COVID-19-Related COPD Exacerbations. *Annals of the American Thoracic Society*. 2026.

Awards, Fellowships, & Grants

2026	Swedish Heart-Lung Foundation Travel Grant , Swedish Heart-Lung Foundation	SEK 12,000
2025	Swedish Heart-Lung Foundation Travel Grant , Swedish Heart-Lung Foundation	SEK 24,000
2025	Wilhelm & Martina Lundgren Science Fund Grant , Stiftelsen Wilhelm och Martina Lundgrens Vetenskapsfond	SEK 25,000
2024	Swedish Heart-Lung Foundation Travel Grant , Swedish Heart-Lung Foundation	SEK 12,000
2024	Sahlgrenska Academy Travel Grant, Iwan & Eleonore Ljunggrens Fond , Sahlgrenska Academy, University of Gothenburg	SEK 8,000
2023	Adlerbertska Foundation Travel Grant , Institute of Medicine, Sahlgrenska Academy, University of Gothenburg	SEK 20,000
2023	Swedish Heart-Lung Foundation Travel Grant , Swedish Heart-Lung Foundation	SEK 20,000
2023	Sahlgrenska Academy Travel Grant, Anna Cederbergs Fond , Sahlgrenska Academy, University of Gothenburg	SEK 10,000
2023	FORTE EpiDem Network Junior Conference Sponsored Attendance , FORTE EpiDem Network	
2022	Adlerbertska Foundation Travel Grant , Institute of Medicine, Sahlgrenska Academy, University of Gothenburg	SEK 20,000
2020	Adlerbertska Foundation Scholarship , Institute of Medicine, Sahlgrenska Academy, University of Gothenburg	SEK 15,000

Presentations

INVITED TALKS & RESEARCH VISITS

- Mar 2025 **Visiting Analyst: Smoking and COVID-19 in the HUNT Data**, Trondheim University Hospital & NTNU, Trondheim, Norway
- Jun 2023 **Optimizing Cox Regression: Comparing Traditional Covariate Adjustment and Modern Propensity Score Methods for Balancing Covariates**, Department of Endocrinology, Sahlgrenska University Hospital
- Oct 2022 **Pharmacoepidemiology and Drug Utilization Research**, 17th Hospital Pharmacists Association of Kenya Conference (Virtual)

SELECTED CONFERENCE PRESENTATIONS

2026. *Why background incidence matters: Sex-, temporal-, and pandemic-related variation in adverse events of special interest shape COVID-19 vaccine safety signals*. Poster presentation, 42nd International Conference on Pharmacoepidemiology & Therapeutic Risk Management (ICPE), Milan, Italy. 29 August–3 September 2026.
2026. *Reproducible smoker's paradox in COVID-19: Comparative evidence from Swedish and Norwegian cohorts*. Poster presentation, Nordic Conference of Epidemiology and Register-Based Health Research (NordicEpi) and Nordic Pharmacoepidemiology (NorPEN) Conference, Tromsø, Norway. 3–5 June 2026.
2025. *Understanding age-specific risk factors for COVID-19: A machine learning approach*. Poster presentation, 41st International Conference on Pharmacoepidemiology & Therapeutic Risk Management (ICPE), Washington, DC, USA. 22–26 August 2025.
2024. *COVID-19 Risk and Effect Modification by Vaccination in Asthma*. Oral presentation, European Respiratory Society (ERS) Congress 2024, Vienna, Austria.
2024. *Risk of Acute Cardiovascular Events in COPD Patients After Experiencing COVID-19-Related vs. Non-COVID-19-Related Exacerbations: A Population-Based Study*. Oral presentation, 40th International Conference on Pharmacoepidemiology & Therapeutic Risk Management (ICPE), Berlin, Germany.
2023. *Smoking Status and COVID-19 Related Risk in Four Swedish Non-Communicable Disease Registers: A Non-Intuitive Association?* Oral presentation, Swedish Register-Based Research Summit, Stockholm, Sweden.
2023. *Impact of Chronic Oral Glucocorticoid Treatment on Mortality in Patients with COVID-19: An Analysis from a Population-Based Cohort*. Poster presentation, 39th International Conference on Pharmacoepidemiology & Therapeutic Risk Management (ICPE), Halifax, Canada.
2023. *The Role of Smoking Status as a Risk Factor in COVID-19 Infection and Severity in a Swedish Population-Based Cohort*. Poster presentation, Swedish Lung Congress 2023, Gothenburg, Sweden.
2022. *Inhaled Corticosteroids and COVID-19 Risk and Outcomes: Findings from a Population-Based Cohort Within the SCIFI-PEARL Project*. Oral presentation, ICPE 2022, Copenhagen, Denmark.
2022. *Characteristics of Asthma Patients with COVID-19 by Disease Severity and Across Age Groups: A Swedish Population-Based Analysis Within the SCIFI-PEARL Project*. Poster presentation, ICPE 2022, Copenhagen, Denmark.
2022. *Characteristics and Risk Factors for COVID-19 Outcomes in a Swedish Population-Based Cohort of COPD Patients from the SCIFI-PEARL Project*. Oral presentation, NordicEPI 2022, Reykjavik, Iceland.
2021. *Characteristics and Outcomes of COVID-19 Patients with Pre-Existing COPD in Sweden: A Population-Based Register Study*. Oral presentation, 13th Nordic Pharmacoepidemiological Network (NorPEN) Meeting (Virtual).

Teaching & Mentoring

University of Gothenburg

Gothenburg, Sweden

RESEARCH AND STATISTICAL SUPERVISOR

2023–2024

- 2024 – Freja Söderbergh (Medical Student), COVID-19 in patients with bronchiectasis.
- 2023 – Omar Noor (Medical Student), COVID-19 in patients with bronchiectasis.
- 2023 – Ágústa Eyjólfssdóttir (Visiting Student, University of Iceland), Predictors of COVID-19 vaccination among immigrants.

Peer Review & Professional Service

PEER REVIEW ACTIVITIES

Ad hoc Journal Reviewer

Therapeutic Advances in Respiratory Disease (2023–)

Journal of Experimental Pharmacology (2024–)

Respiratory Research (2024–)

CHEST (2024–)

Journal of Diabetes & Metabolic Disorders (2026–)

Other Editorial Roles

Abstract Reviewer, International Society for Pharmacoepidemiology (ISPE) Annual Meeting (Halifax 2023; Berlin 2024; Washington, DC 2025; Milan 2026).

SCIENTIFIC SOCIETIES

International Society for Pharmacoepidemiology (ISPE), Member (2021–Present)

Nordic Pharmacoepidemiologic Network (NorPEN), Member (2022–Present)

BREATHE Sweden Consortium (2022–Present)

Observational Health Data Sciences and Informatics (OHDSI), Member (2023–Present)

Swedish Epidemiological Society (SVEP), Member (2024–Present)